Lecture 1: Electric Force and Charge

1. Introduction

Have you ever rubbed a balloon on your hair and then noticed that your hair sticks to it?

Or:

Have you ever received a small electric shock after touching a metal object?

Why does this happen?

The answer is related to **electric charge**.

Today we are going to understand what electric charge is, where it comes from, and how charged objects interact with each other.

2. What is Electric Charge?

Simple definition:

Electric charge is a physical property of matter that causes objects to experience electrical forces.

Charge is represented by the symbol:

q

The SI unit of charge is:

Coulomb (C)

For example:

$q = 5\ C$ means the object has a charge of 5 Coulombs.

3. Where Does Electric Charge Come From?

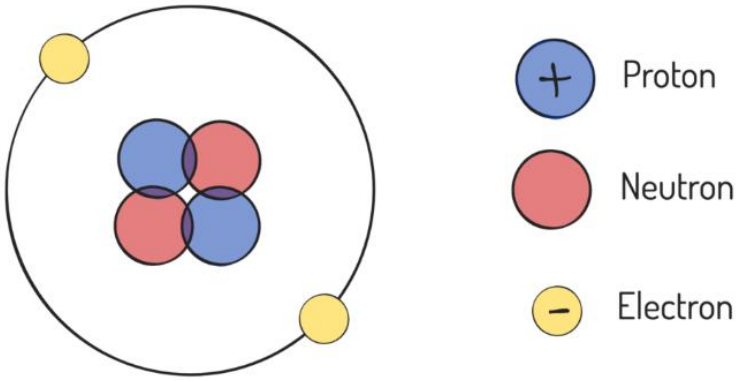
Everything around us is made up of **atoms**.

An atom contains:

- **Protons**
- **Neutrons**
- **Electrons**

Simple structure:

Particle	Charge
Proton	Positive (+)
Electron	Negative (−)
Neutron	No charge



Remember:

Proton → +

Electron → −

Neutron → 0

The positive and negative charges come from protons and electrons.

4. What Makes an Object Charged?

Normally, an object has approximately equal numbers of:

- Protons
- Electrons

Therefore, the object is **electrically neutral**.

Neutral object:

Number of protons = Number of electrons

But electrons can move from one object to another.

If an object gains electrons:

It becomes:

NEGATIVELY CHARGED

If an object loses electrons:

It becomes:

POSITIVELY CHARGED

Very important:

Electrons move; protons normally do not move from one object to another.

5. Easy Example - Balloon

Suppose you rub a balloon against your hair.

Some electrons move from one material to another.

The balloon gains electrons.

Therefore:

Balloon → Negative charge

Your hair loses electrons.

Therefore:

Hair → Positive charge

This is called **charging by friction**.

This is why your hair may stand up or stick to the balloon.

6. Types of Electric Charge

There are two types of electric charge:

Positive Charge (+)

Caused by a lack of electrons.

Negative Charge (−)

Caused by an excess of electrons.

There is also:

Neutral

No net electric charge.

7. The Most Important Rule of Charge

There is a very simple rule:

Like charges repel.

Positive + Positive → **Repel**

Negative + Negative → **Repel**

Unlike charges attract.

Positive + Negative → **Attract**

You can remember this:

Like → Repel
Unlike → Attract

Note: Memorize this because it will be used throughout the course.

8. What Does "Force" Mean?

Before understanding **electric force**, let's understand force.

Simple definition:

Force is a push or pull that can change the motion or shape of an object.

Examples:

- Pushing a door
- Pulling a chair
- Kicking a football
- Gravity pulling an object downward

The SI unit of force is:

Newton (N)

9. What is Electric Force?

Now combine the two ideas.

Electric Force

Electric force is the force of attraction or repulsion between charged objects.

For example:

Positive and Positive

They repel each other.

(+) ← → (+)

Negative and Negative

They repel each other.

(−) ← → (−)

Positive and Negative

They attract each other.

(+) → ← (−)

10. What Determines Electric Force?

The electric force between two charges depends mainly on:

1. Amount of charge

Larger charges → **larger force**

2. Distance between charges

Greater distance → **smaller force**

So:

More charge = More electric force

More distance = Less electric force

This leads us to **Coulomb's Law**.

11. Coulomb's Law

Coulomb's Law tells us how to calculate the electric force between two point charges.

Formula:

$$F = k\frac{|q_1 q_2|}{r^2}$$

Where:

F = Electric force, measured in Newton's (N)

q_1 = First charge, measured in Coulombs (C)

q_2 = Second charge, measured in Coulombs (C)

r = Distance between the charges, measured in meters (m)

k = Coulomb's constant

$$k = 9 \times 10^9 \, Nm^2/C^2$$

12. Understanding the Formula without Fear

Don't memorize the formula blindly.

Look at what it tells us:

$$F = k\frac{q_1 q_2}{r^2}$$

Charges are on top:

More charge → **More force**

Distance is at the bottom:

More distance → **Less force**

Distance is squared:

This is very important.

If distance becomes **2 times larger**:

$$F \to \frac{1}{2^2}$$

$$F \to \frac{1}{4}$$

So the force becomes **one-fourth**.

13. Simple Numerical Example

Suppose:

$$q_1 = 2C$$

$$q_2 = 3C$$

And:

$$r = 2m$$

Find the electric force.

Step 1: Write the formula

$$F = k\frac{q_1 q_2}{r^2}$$

Coulomb's constant, represented by k.

In Coulomb's Law: the constant k has a standard value: 9×10^9

So when you substitute values, you replace k with 9×10^9

Step 2: Substitute values

$$F = 9 \times 10^9 \frac{(2)(3)}{(2)^2}$$

Step 3:

$$F = 9 \times 10^9 \frac{6}{4}$$

$$F = 13.5 \times 10^9$$

Therefore:

$$F = 1.35 \times 10^{10}\ N$$

14. A More Realistic Example (Micro charges)

Real-world charges are often much smaller than 1 Coulomb.

For example:

$$q_1 = 2\mu C$$

$$q_2 = 3\mu C$$

Remember:

$$1\mu C = 10^{-6} C$$

The symbol **μ** is called **"mu"** (pronounced like **"myoo"**).

Suppose:

$$r = 0.5m$$

Then:

$$F = k\frac{q_1 q_2}{r^2}$$

First convert:

$$2\mu C = 2 \times 10^{-6} C$$

$$3\mu C = 3 \times 10^{-6} C$$

Then substitute into the formula.

This introduces us to the idea that **units must be converted before using formulas**.

15. Why Does Distance Matter So Much?

This is called the **inverse-square relationship**.

Coulomb's Law contains:

$$\frac{1}{r^2}$$

This means force decreases very quickly as distance increases.

Example:

If distance is:

r

Force = **F**

If distance becomes:

2r

Force becomes:

$$\frac{F}{4}$$

If distance becomes:

3r

Force becomes:

$$\frac{F}{9}$$

If distance becomes:

4r

Force becomes:

$$\frac{F}{16}$$

Easy way:

Double the distance → Force becomes 1/4

Triple the distance → Force becomes 1/9

16. Electric Force vs Gravitational Force

Question: Is electric force similar to gravity?

Yes, in some ways.

Gravity: Acts between masses.

Electric force: Acts between charges.

Gravity	Electric Force
Depends on mass	Depends on charge
Always attractive	Attractive or repulsive
Very important for large objects	Very important at atomic/electronic scales

Important difference:

Gravity:

Only attracts

Electric force:

Can attract or repel (push away from each other)

17. Electric Charge in Technology

Now connect the topic to Computer Science.

Computers: Computers work using electrical signals.

Electronic circuits: Circuits depend on the movement of electric charges.

Transistors: Transistors control electrical signals and are fundamental to processors.

Batteries: Batteries create a potential difference that drives electric current through circuits.

Touchscreens: Touchscreens detect changes in electrical properties when your finger touches the screen.

Static electricity: Static charge can sometimes damage sensitive electronic components.

Note: So the charge we are studying today is not just theoretical. It is part of the technology you use every day.

18. Charging by Friction

One simple method of charging an object is:

Charging by friction

When two different materials are rubbed together, electrons may transfer from one material to another.

Examples:

- Balloon + hair
- Comb + hair
- Plastic + cloth

The object gaining electrons becomes: **Negative**

The object losing electrons becomes: **Positive**

19. Quick Classroom Questions:

Q1. What are the three main particles in an atom?

Q2. Which particle has negative charge?

Q3. Which particle has positive charge?

Q4. What happens between two positive charges?

Q5. What happens between positive and negative charges?

Q6. What happens to electric force when distance increases?

Q7. What is the SI unit of charge?

Q8. What is the SI unit of force?

Answers:

Q1. What are the three main particles in an atom?

Protons, neutrons and electrons.

Q2. Which particle has negative charge?

Electron.

Q3. Which particle has positive charge?

Proton.

Q4. What happens between two positive charges?

Repulsion.

Q5. What happens between positive and negative charges?

Attraction.

Q6. What happens to electric force when distance increases?

It decreases.

Q7. What is the SI unit of charge?

Coulomb (C).

Q8. What is the SI unit of force?

Newton (N).

20. One-Minute Student Challenge

A

+ +

B

− −

C

+ −

Tell:

"Attract or repel?"

Answers:

A → Repel

B → Repel

C → Attract

"What if I double the distance between the charges?"

Answer: **Electric force becomes one-fourth.**

21. Final Summary

Electric Charge

Electric charge is a property of matter.

There are two types:

Positive (+)

Negative (−)

Particles

Proton → Positive

Electron → Negative

Neutron → Neutral

Charge interaction

Like charges → Repel

Unlike charges → Attract

Electric Force

Electric force is the attraction or repulsion between charged objects.

Coulomb's Law

$$F = k\frac{|q_1 q_2|}{r^2}$$

Remember:

More charge → More force

More distance → Less force

Double distance → One-fourth force

TODAY'S PHYSICS SURVIVAL RULES

Charge:
+ Positive | **−** Negative

Remember:

LIKE → REPEL

UNLIKE → ATTRACT

And don't forget:

$$F = k\frac{|q_1 q_2|}{r^2}$$

More charge → More force

More distance → Less force

Next lecture:

Electric Field

And yes… there will be more formulas.